

### Problem 13.6.1

Show that

a)  $\sinh z = \sinh x \cos y + i \cosh x \sin y$

b)  $\cosh z = \cosh x \cos y + i \sinh x \sin y$

### Solution

a)

$$\begin{aligned}
 \sinh z &= \frac{1}{2}(e^z - e^{-z}) \\
 &= \frac{1}{2}(e^{x+iy} - e^{-x-iy}) \\
 &= \frac{1}{2}(e^x e^{iy} - e^{-x} e^{-iy}) \\
 &= \frac{1}{2}(e^x (\cos y + i \sin y) - e^{-x} (\cos y - i \sin y)) \\
 &= \frac{1}{2}(e^x \cos y + i e^x \sin y - e^{-x} \cos y + i e^{-x} \sin y) \\
 &= \frac{1}{2}((e^x - e^{-x}) \cos y + i(e^x + e^{-x}) \sin y) \\
 &= \boxed{\sinh x \cos y + i \cosh x \sin y}
 \end{aligned}$$

b)

$$\begin{aligned}
 \cosh z &= \frac{1}{2}(e^z + e^{-z}) \\
 &= \frac{1}{2}(e^{x+iy} + e^{-x-iy}) \\
 &= \frac{1}{2}(e^x e^{iy} + e^{-x} e^{-iy}) \\
 &= \frac{1}{2}(e^x (\cos y + i \sin y) + e^{-x} (\cos y - i \sin y)) \\
 &= \frac{1}{2}(e^x \cos y + i e^x \sin y + e^{-x} \cos y - i e^{-x} \sin y) \\
 &= \frac{1}{2}((e^x + e^{-x}) \cos y + i(e^x - e^{-x}) \sin y) \\
 &= \boxed{\cosh x \cos y + i \sinh x \sin y}
 \end{aligned}$$

### Problem 13.6.3

Show that

a)  $\cosh^2 z - \sinh^2 z = 1$

b)  $\cosh^2 z + \sinh^2 z = \cosh 2z$

**Solution**

a)

$$\begin{aligned}
 \cosh^2 z - \sinh^2 z &= \frac{1}{4}(e^z + e^{-z})^2 - \frac{1}{4}(e^z - e^{-z})^2 \\
 &= \frac{1}{4}[(e^{2z} + 2 + e^{-2z}) - (e^{2z} - 2 + e^{-2z})] \\
 &= \boxed{1}
 \end{aligned}$$

b)

$$\begin{aligned}
 \cosh^2 z + \sinh^2 z &= \frac{1}{4}(e^z + e^{-z})^2 + \frac{1}{4}(e^z - e^{-z})^2 \\
 &= \frac{1}{4}[(e^{2z} + 2 + e^{-2z}) + (e^{2z} - 2 + e^{-2z})] \\
 &= \frac{1}{4}(2e^{2z} + 2e^{-2z}) \\
 &= \frac{1}{2}(e^{2z} + e^{-2z}) \\
 &= \boxed{\cosh 2z}
 \end{aligned}$$

**Problem 13.6.7**Find, in the form  $u + iv$ a)  $\cos i$ b)  $\sin i$ **Solution**

a)

$$\cos i = \boxed{\cosh 1}$$

b)

$$\sin i = \boxed{i \sinh 1}$$

**Problem 13.6.13**

Using the definitions, prove

a)  $\cos z$  is even,  $\cos(-z) = \cos z$ b)  $\sin z$  is odd,  $\sin(-z) = -\sin z$ **Solution**

a)

$$\begin{aligned}
 \cos(-z) &= \frac{1}{2}(e^{i \cdot (-z)} + e^{-i \cdot (-z)}) \\
 &= \frac{1}{2}(e^{-iz} + e^{iz}) \\
 &= \boxed{\cos z}
 \end{aligned}$$

b)

$$\begin{aligned}
 \sin(-z) &= \frac{1}{2i} \left( e^{i \cdot (-z)} - e^{-i \cdot (-z)} \right) \\
 &= \frac{1}{2i} (e^{-iz} - e^{iz}) \\
 &= \boxed{-\sin z}
 \end{aligned}$$

**Problem 13.6.16**

Find all solutions

$$\sin z = 100$$

**Solution**

$$\sin z = \frac{1}{2i} (e^{iz} - e^{-iz}) = 100$$

Let  $w = e^{iz}$ . Then  $w - w^{-1} = 100i$ 

$$w^2 - 200iw - 1 = 0 \quad \implies \quad w = i(100 \pm \sqrt{9999})$$

Thus  $e^{iz} = i(100 \pm \sqrt{9999})$ , so

$$\begin{aligned}
 iz &= \ln(i(100 \pm \sqrt{9999})) \\
 &= \ln|i(100 \pm \sqrt{9999})| + i \operatorname{Arg}(i(100 \pm \sqrt{9999})) + i2\pi n \\
 &= \ln(100 \pm \sqrt{9999}) + i\frac{\pi}{2} + i2\pi n \\
 &= \pm \ln(100 + \sqrt{9999}) + i\pi \left( 2n + \frac{1}{2} \right)
 \end{aligned}$$

Dividing by  $i$ ,

$$z = \pi \left( 2n + \frac{1}{2} \right) \pm \ln(100 + \sqrt{9999}) \quad n \in \mathbb{Z}$$

**Problem 13.7.5**

Find

$$\operatorname{Ln}(-11)$$

**Solution**

$$\begin{aligned}
 \operatorname{Ln}(-11) &= \ln|-11| + i \operatorname{Arg}(-11) \\
 &= \boxed{\ln 11 + i\pi}
 \end{aligned}$$

**Problem 13.7.7**

Find

$$\operatorname{Ln}(4 - 4i)$$

**Solution**

$$\begin{aligned}\operatorname{Ln}(4 - 4i) &= \ln |4 - 4i| + i \operatorname{Arg}(4 - 4i) \\ &= \ln \sqrt{4^2 + 4^2} + i \tan^{-1} \left( \frac{-4}{4} \right) \\ &= \ln \sqrt{32} + i \tan^{-1}(-1) \\ &= \boxed{\frac{1}{2} \ln 32 - i \frac{\pi}{4}}\end{aligned}$$

**Problem 13.7.8**

Find

$$\operatorname{Ln}(1 \pm i)$$

**Solution**

$$\begin{aligned}\operatorname{Ln}(1 \pm i) &= \ln |1 \pm i| + i \operatorname{Arg}(1 \pm i) \\ &= \ln \sqrt{2} + i \tan^{-1} \left( \frac{\pm 1}{1} \right) \\ &= \ln \sqrt{2} + i \tan^{-1}(\pm 1) \\ &= \boxed{\frac{1}{2} \ln 2 \pm i \frac{\pi}{4}}\end{aligned}$$

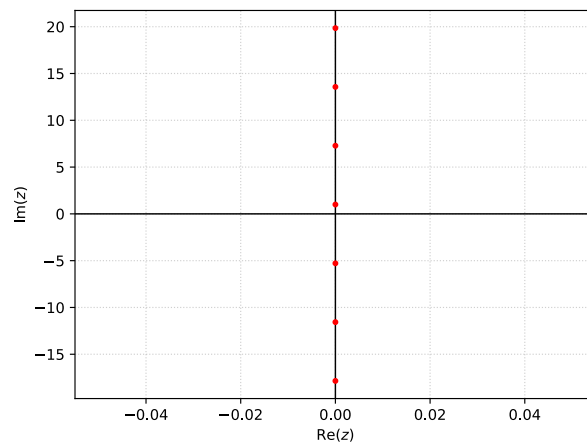
**Problem 13.7.15**

Find all values and graph some in the complex plane

$$\ln(e^i)$$

**Solution**

$$\begin{aligned}\ln(e^i) &= \ln |e^i| + i \operatorname{Arg}(e^i) + i2\pi n \\ &= \ln 1 + i + i2\pi n \\ &= \boxed{i(1 + 2\pi n) \quad n \in \mathbb{Z}}\end{aligned}$$

Figure 1: Solutions of  $\ln(e^i)$ **Problem 13.7.19**Solve for  $z$ 

$$\ln z = 4 - 3i$$

**Solution**

$$\begin{aligned}
 e^{\ln z} &= e^{4-3i} \\
 z &= e^{4-3i} \\
 &= e^4 (\cos(-3) + i \sin(-3)) \\
 &= \boxed{e^4 (\cos 3 - i \sin 3)}
 \end{aligned}$$

**Problem 13.7.23**

Find the principal value.

$$(1+i)^{1-i}$$

**Solution**

$$\begin{aligned}
(1+i)^{1-i} &= \exp((1-i) \operatorname{Ln}(1+i)) \\
&= \exp\left((1-i) \left(\frac{1}{2} \ln 2 + i \frac{\pi}{4}\right)\right) && \text{(from 13.7.8)} \\
&= \exp\left((1-i) \left(\frac{1}{2} \ln 2 + i \frac{\pi}{4}\right)\right) \\
&= \exp\left(\frac{1}{2} \ln 2 + i \frac{\pi}{4} - i \frac{1}{2} \ln 2 + \frac{\pi}{4}\right) \\
&= \exp\left(\frac{\pi}{4} + \frac{1}{2} \ln 2\right) \exp\left(i \left(\frac{\pi}{4} - \frac{1}{2} \ln 2\right)\right) \\
&= \boxed{\exp\left(\frac{\pi}{4} + \frac{1}{2} \ln 2\right) \left[\cos\left(\frac{\pi}{4} - \frac{1}{2} \ln 2\right) + i \sin\left(\frac{\pi}{4} - \frac{1}{2} \ln 2\right)\right]}
\end{aligned}$$

**Problem 14.1.1**

Find and sketch the path

$$z(t) = \left(1 + \frac{1}{2}i\right)t \quad (2 \leq t \leq 5)$$

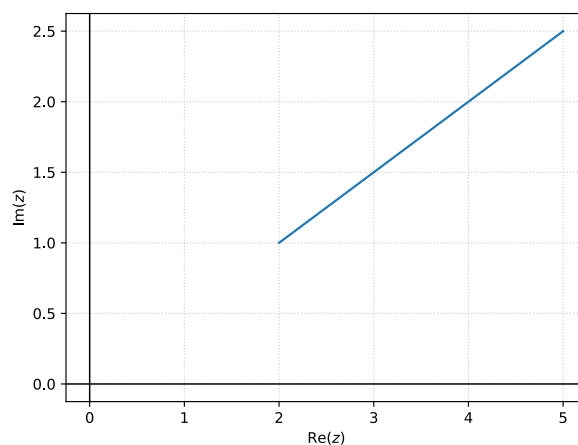
**Solution**

Figure 2:  $z(t)$

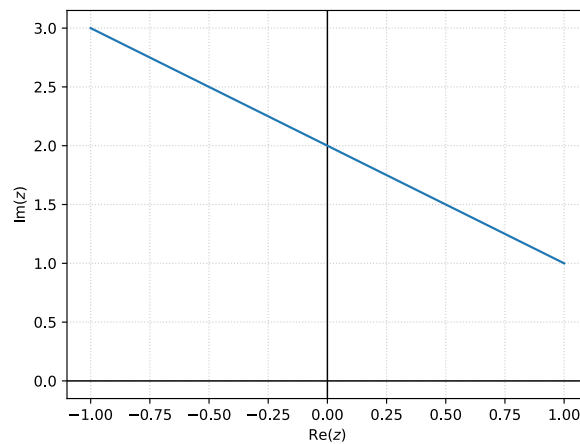
**Problem 14.1.11**

Find a parametric representation and sketch the path

Segment from  $(-1, 1)$  to  $(1, 3)$

**Solution**

$$z(t) = (t-1) + i(t+1) \quad t \in [0, 2]$$

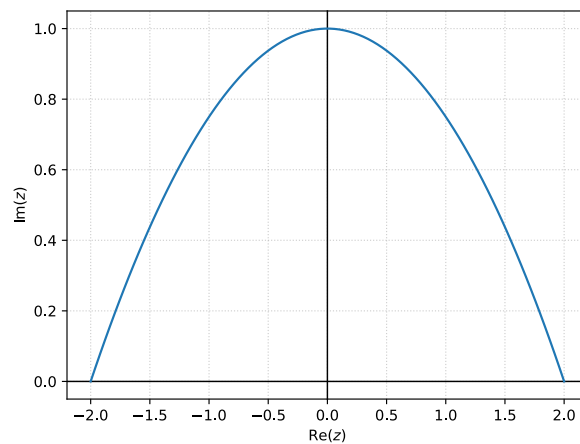
Figure 3:  $z(t)$ **Problem 14.1.19**

Find a parametric representation and sketch the path

$$\text{Parabola } y = 1 - \frac{1}{4}x^2 \quad (-2 \leq x \leq 2)$$

**Solution**

$$z(t) = t + i \left( 1 - \frac{1}{4}t^2 \right) \quad t \in [-2, 2]$$

Figure 4:  $z(t)$ **Problem 14.1.21**

Integrate by the first method or state why it does not apply and use the second method. Show the details.

$$\int_C \operatorname{Re} z \, dz, \quad C \text{ the shortest path from } 1 + i \text{ to } 3 + 3i$$

**Solution**

$$f(z) = \operatorname{Re} z = u(x, y) + iv(x, y)$$

$$\begin{aligned} u &= x & v &= 0 \\ u_x &= 1 & v_x &= 0 \\ u_y &= 0 & v_y &= 0 \end{aligned}$$

$u_x \neq v_y$  therefore  $f$  is not analytic and the first method is invalid.

$$\int_C f(z) \, dz = \int_a^b f[z(t)] \dot{z}(t) \, dt$$

$$\begin{aligned} z(t) &= t + it & t &\in [1, 3] \\ \dot{z}(t) &= 1 + i \end{aligned}$$

$$\begin{aligned} \int_C f(z) \, dz &= \int_1^3 f[z(t)] \dot{z}(t) \, dt \\ &= \int_1^3 t \cdot (1 + i) \, dt \\ &= (1 + i) \frac{1}{2} t^2 \Big|_1^3 \\ &= \boxed{4 + 4i} \end{aligned}$$

**Problem 14.1.25**

$$\int_C z \exp(z^2) \, dz, \quad C \text{ from } 1 \text{ along the axes to } i$$

**Solution**

**Problem 14.1.29**

$$\int_C \operatorname{Im} z^2 \, dz \quad \text{counterclockwise around the triangle with vertices } 0, 1, i$$

**Solution**